

GNCAirstrike: Longitudinal Missile Guidance, Navigation and Control

A Short-Period Autopilot with Kalman-Filtered State Estimation, LQR Control, and
Proportional Navigation Guidance

Meet

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Scope Statement

This project models the **longitudinal, short-period dynamics** of a missile only (angle of attack α and pitch rate q). Phugoid dynamics and lateral-directional motion are explicitly out of scope; this is a deliberate scoping decision to allow depth in guidance-law design and statistical validation rather than breadth in vehicle dynamics. A simple terrain-proximity safety override is retained as a secondary feature and is not the guidance law of the system.

1 System Model

1.1 Short-Period Longitudinal Dynamics

The airframe is modeled as a linear time-invariant state-space system with states angle of attack α and pitch rate q , control input the commanded fin deflection δ_c , and outputs normal acceleration A_z and pitch rate q :

$$\dot{\mathbf{x}} = A\mathbf{x} + Bu, \quad \mathbf{y} = C\mathbf{x} + Du \quad (1)$$

with state vector, input, and output

$$\mathbf{x} = \begin{bmatrix} \alpha \\ q \end{bmatrix}, \quad u = \delta_c, \quad \mathbf{y} = \begin{bmatrix} A_z \\ q \end{bmatrix} \quad (2)$$

and the identified plant matrices

$$A = \begin{bmatrix} -1.064 & 1.000 \\ 290.26 & 0.00 \end{bmatrix}, \quad B = \begin{bmatrix} -0.25 \\ -331.40 \end{bmatrix}, \quad C = \begin{bmatrix} -123.34 & 0.00 \\ 0.00 & 1.00 \end{bmatrix}, \quad D = \begin{bmatrix} -13.51 \\ 0 \end{bmatrix}. \quad (3)$$

The open-loop poles of the q/δ_c transfer function determine the natural short-period frequency and damping of the airframe prior to any control action.

2 Navigation: Kalman State Estimation

Since α is not directly measurable in practice, a Kalman filter estimates the full state $\hat{\mathbf{x}}$ from noisy outputs. The plant is augmented with process noise w and measurement noise v :

$$\dot{\mathbf{x}} = A\mathbf{x} + Bu + Gw, \quad \mathbf{y} = C\mathbf{x} + Du + Hv, \quad (4)$$

with $G = I_2$, $H = 0$, and noise covariances

$$\bar{Q} = \text{diag}(1.5 \times 10^{-4}, 1.5 \times 10^{-4}), \quad \bar{R} = \text{diag}(0.55, 0.55). \quad (5)$$

The steady-state Kalman gain L is obtained from the filter algebraic Riccati equation, and the estimator dynamics are

$$\dot{\hat{\mathbf{x}}} = A\hat{\mathbf{x}} + Bu + L(\mathbf{y} - C\hat{\mathbf{x}} - Du). \quad (6)$$

Estimator stability is confirmed by the eigenvalues of the observer error dynamics matrix

$$A_{ob} = A - LC, \quad (7)$$

which must lie strictly in the left half-plane. Process and measurement noise are sampled at $\Delta T_1 = 0.75 \text{ s}$ and $\Delta T_2 = 0.25 \text{ s}$ respectively.

3 Control: LQR Inner Loop

The inner loop regulates the estimated state $\hat{\mathbf{x}}$ toward a commanded reference by minimizing the quadratic cost

$$J = \int_0^\infty (\mathbf{x}^\top Q \mathbf{x} + u^\top R u) dt, \quad Q = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix}, \quad R = 0.5, \quad (8)$$

which yields the state-feedback control law

$$u = -K\hat{\mathbf{x}}, \quad (9)$$

with K the solution of the continuous algebraic Riccati equation

$$A^\top S + SA - SBR^{-1}B^\top S + Q = 0, \quad K = R^{-1}B^\top S. \quad (10)$$

The closed-loop plant matrix is $A_{cl} = A - BK$, and closed-loop stability is verified via $\text{eig}(A_{cl})$.

3.1 Outer Loop: Flight-Path-Angle Tracking

A proportional (P-only) controller closes an outer loop around flight path angle γ :

$$q_{cmd}(t) = K_P(\gamma_{cmd}(t) - \gamma(t)), \quad K_P = -1.0, \quad (11)$$

which supplies the pitch-rate reference tracked by the inner LQR loop. (Integral and derivative gains are currently zero; this is a proportional-only outer loop, not a full PID, despite using Simulink's PID block.)

4 Guidance

4.1 Legacy Guidance Logic (Baseline)

The current implementation computes flight-path-angle commands from great-circle geometry rather than from a closed-loop guidance law. Great-circle distance between missile position (ϕ_1, λ_1) and target position (ϕ_2, λ_2) uses the Haversine formula:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos\phi_1 \cos\phi_2 \sin^2\left(\frac{\Delta\lambda}{2}\right), \quad c = 2 \text{atan2}(\sqrt{a}, \sqrt{1-a}), \quad d = R_\oplus c, \quad (12)$$

with $R_{\oplus} = 6371$ km. The commanded flight path angle is then the static geometric angle to the target,

$$\gamma_{cmd} = \text{atan}\left(\frac{|h_{target} - h_0|}{d}\right), \quad (13)$$

overridden to a fixed 5° climb whenever range to the obstacle drops below a 2 km threshold:

$$\gamma_{cmd} \leftarrow \gamma_{override} = 5^\circ \quad \text{if } d_{obstacle} < 2 \text{ km}. \quad (14)$$

This produces a correct terminal geometry but is not reactive to engagement dynamics — it computes a fixed target bearing once and only reacts to the obstacle via a hard, non-smooth trigger.

4.2 Proportional Navigation Guidance (Planned Replacement)

Proportional Navigation (PN) replaces the static FPA command above the terrain-safety layer. Let R_{TM} be the missile-to-target range vector, λ the line-of-sight (LOS) angle, and V_c the closing velocity. PN commands a lateral (here, normal) acceleration proportional to LOS rotation rate:

$$a_{cmd} = N' V_c \dot{\lambda}, \quad (15)$$

where N' is the navigation constant (typically 3–5) and $\dot{\lambda}$ is computed from successive LOS angle samples,

$$\lambda = \text{atan2}(y_T - y_M, x_T - x_M), \quad V_c = -\frac{d}{dt}\|\mathbf{R}_{TM}\|. \quad (16)$$

The commanded acceleration a_{cmd} is converted to the same A_z reference currently used by the inner LQR loop, so PN drops in as a direct replacement for the outer-loop FPA command with no change to the navigation or control layers.

4.3 Terrain / Obstacle Safety Override (Retained, Unchanged)

The existing hard override — forcing a fixed climb angle when within 2 km of the obstacle/terrain threshold — is retained as-is as a secondary safety layer beneath the PN guidance law. It is not represented as part of the guidance law itself and is described in this report as a safety feature, not a path-planning or avoidance algorithm.

5 Monte Carlo Validation Methodology

5.1 Randomized Parameters

Each of N independent trials redraws the following quantities:

- **Engagement geometry:** initial position ($\text{LAT}_{init}, \text{LON}_{init}$) and/or target position ($\text{LAT}_{target}, \text{LON}_{target}$), perturbed uniformly within a realistic radius of their nominal values.
- **Process and measurement noise:** independent realizations of $w \sim \mathcal{N}(0, \bar{Q})$ and $v \sim \mathcal{N}(0, \bar{R})$ per trial, using a distinct random seed.
- **Initial condition offset:** small random initial (α_0, q_0) instead of a fixed trim start.

5.2 Metrics Collected per Trial

$$\text{Miss distance: } m_i = \|\mathbf{x}_{target} - \mathbf{x}_{missile}(t_f)\|_i \quad (17)$$

$$\text{Estimation error: } e_i(t) = \hat{\mathbf{x}}_i(t) - \mathbf{x}_i(t) \quad (18)$$

$$\text{Control saturation flag: } s_i = \mathbb{K}\{|\delta_c(t)| \geq \delta_{max} \text{ for any } t\} \quad (19)$$

$$\text{Terrain-override trigger flag: } o_i = \mathbb{K}\{\gamma_{override} \text{ fired during trial } i\} \quad (20)$$

5.3 Aggregate Statistics

Across all N trials, the sample mean and standard deviation of miss distance,

$$\bar{m} = \frac{1}{N} \sum_{i=1}^N m_i, \quad \sigma_m = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (m_i - \bar{m})^2}, \quad (21)$$

the success rate against a defined miss-distance threshold m_{th} ,

$$P_{success} = \frac{1}{N} \sum_{i=1}^N \mathbb{K}\{m_i \leq m_{th}\}, \quad (22)$$

and the Circular Error Probable (radius containing 50% of trials, computed from the empirical order statistics of $\{m_i\}$),

$$\text{CEP}_{50} = m_{(\lceil 0.5N \rceil)} \quad (\text{sorted ascending}). \quad (23)$$

5.4 Execution Strategy for Speed

To keep $N \gtrsim 200$ –500 trials tractable across multiple randomized parameters, trials are executed with MATLAB’s `parsim` using an array of `Simulink.SimulationInput` objects (one per trial, each carrying its own randomized parameter set) rather than a serial `for`-loop calling `sim()` repeatedly. This distributes trials across available workers via the Parallel Computing Toolbox, and results are logged directly to arrays rather than to per-trial `.mat` files to avoid I/O overhead.

6 Summary

GNC Layer	Method	Status
Navigation	Kalman filter (steady-state)	Implemented
Control	LQR inner loop + P outer loop	Implemented
Guidance	Static LOS/FPA command	Baseline; being replaced
Guidance	Proportional Navigation	Planned addition
Safety	Terrain-proximity FPA override	Retained, unchanged
Validation	Monte Carlo (geometry + noise)	Planned addition